Department of Artificial Intelligence

భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
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Reinforcement Learning (AI3000)

Lecture 01

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Recap: Markov Chains

History $X_0, \ldots, X_{n-1}$ → **Present** X_n → **Next state** X_{n+1}

Once the present state is known, the history provides no additional information about the next state.

Markov Chain

Fix a discrete set $\mathcal{X}$

Definition (Markov Property)

A discrete-time process $\{X_n\}_{n\geq 0} = \{X_0, X_1, \dots\}$ taking values in $\mathcal{X}$ is called a Markov chain if:

for every $n \geq 0$, every feasible history $(x_0, \dots, x_n) \in \mathcal{X}^{n+1}$, and every $y \in \mathcal{X}$,

$$\mathbb{P}(\{X_{n+1} = y\} \mid \{X_0 = x_0, \dots, X_n = x_n\}) \;=\; \mathbb{P}(\{X_{n+1} = y\} \mid \{X_n = x_n\}).$$

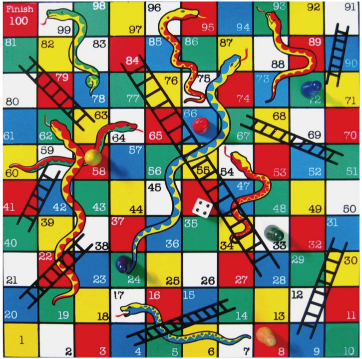
Equivalent conditional-independence view

$$\sigma(X_0, \dots, X_{n-1}) \perp\!\!\!\perp \sigma(X_{n+1}, X_{n+2}, \dots) \,\Big|\, X_n.$$

Thus, conditioned on the present, the future is independent of the past.

The set $\mathcal{X}$ is called the **state space**.

Example: Snakes and Ladders



- X_n: player's square after move n
- $\mathcal{X} = \{1, \dots, 100\}$.
- From $X_n = 6$: $X_{n+1} \in \{7, 8, 31, 10, 11, 12\}$.
- A roll of 3 lands on square 9; the ladder moves the player to square 31.

For every feasible history ending at square 6 and every $x \in \mathcal{X}$,

$$\mathbb{P}(\{X_{n+1} = x\} \mid \{X_0 = x_0, \dots, X_{n-1} = x_{n-1}, X_n = 6\}) = \mathbb{P}(\{X_{n+1} = x\} \mid \{X_n = 6\}) .$$

Transition Probability Matrix (TPM)

Definition (Transition Probability Matrix)

For a DTMC $\{X_n\}_{n\geq 0}$ on a finite state space $\mathcal{X}$, its one-step transition probability matrix (TPM) at time n is

$$P^{(n)}(y \mid x) \;:=\; \mathbb{P}(\{X_{n+1} = y\} \mid \{X_n = x\}), \qquad x, y \in \mathcal{X}.$$

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Each row is a probability mass function:

$$P^{(n)}(y \mid x) \geq 0, \qquad \sum_{y\in\mathcal{X}} P^{(n)}(y \mid x) = 1.$$

Hence $P^{(n)}$ is **row stochastic**, and

$$P^{(n)}\mathbf{1} = \mathbf{1}.$$

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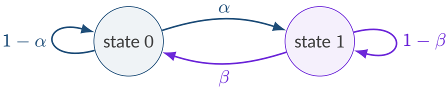

$$P \;=\; \begin{array}{c c} & \begin{array}{cc} 0 & \;\;1 \end{array} \\ \begin{array}{c} 0 \\ 1 \end{array} & \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix} \end{array}.$$

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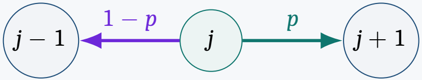
Example: Biased Random Walk

- $Z_1, Z_2, \ldots$ are i.i.d., with
$$\mathbb{P}(\{Z_t = +1\}) = p, \qquad \mathbb{P}(\{Z_t = -1\}) = 1 - p.$$
- **Position:** $S_0 = 0$ and $S_n = S_0 + \sum_{t=1}^{n} Z_t$
- The process $\{S_n\}_{n \geq 0}$ is called a **random walk**
- State space: $\mathcal{X} = \mathbb{Z}$, the set of integers

$$\mathbb{P}(\{S_{n+1} = k\} \mid \{S_0 = s_0, \ldots, S_n = j\}) \;=\; \mathbb{P}(\{S_{n+1} = k\} \mid \{S_n = j\}) \;=\; P(k \mid j).$$

One step from state j



One-step transition kernel

$$P(k \mid j) \;=\; \begin{cases} p, & k = j+1, \\ 1-p, & k = j-1, \\ 0, & \text{otherwise.} \end{cases}$$

Markov Decision Process

Henceforth, all random variables are defined on a common probability space $(\Omega, \mathscr{F}, \mathbb{P})$.

Markov Decision Process (MDP)

- An MDP is a generalization of a Markov process (or Markov chain) with the following elements:
  - **State space** $\mathcal{S}$
  - **Action space** $\mathcal{A}$
  - **Reward alphabet** $\mathcal{R} \subseteq \mathbb{R}$
- A process $\{(S_n)_{n\geq 0}, (A_n)_{n\geq 0}, (R_n)_{n\geq 1}\}$ is called a Markov decision process if:
  - $S_0 \xrightarrow{A_0} (S_1, R_1) \xrightarrow{A_1} (S_2, R_2) \xrightarrow{A_2} (S_3, R_3) \cdots$
  - R_{n+1} is a deterministic or randomised function of (S_n, A_n, S_{n+1}) for all $n \geq 0$
  - Probabilistically, for all feasible trajectories of states, actions, and rewards,

$$\mathbb{P}\big(\{S_{n+1} = s', \ R_{n+1} = r\} \mid \{S_0 = s_0, \ A_0 = a_0, \ldots, S_n = s, \ A_n = a\}\big)$$
$$= \ \mathbb{P}\big(\{S_{n+1} = s', \ R_{n+1} = r\} \mid \{S_n = s, \ A_n = a\}\big).$$

  - The state S_n captures all information about the history $(S_0, A_0, S_1, R_1, \ldots, S_{n-1}, A_{n-1}, S_n, R_n)$
- In general, the choice of A_n could be based on the entire history up to time n

Time-Homogeneous MDP

- If the probability
$$\mathbb{P}\big(\{S_{n+1} = s',\ R_{n+1} = r\} \mid \{S_n = s,\ A_n = a\}\big)$$
is not a function of n, then the MDP is said to be **time-homogeneous**
- A state-reward dynamics in a time-homogeneous MDP may be specified by a **transition matrix**
$$P \;=\; \big\{P(s', r \mid s, a) :\ (s, a, s', r) \in \mathcal{S} \times \mathcal{A} \times \mathcal{S} \times \mathcal{R}\big\}.$$
- As usual, we have the stochastic structure
$$\sum_{s' \in \mathcal{S}} \sum_{r \in \mathcal{R}} P(s', r \mid s, a) \;=\; 1 \qquad \forall\,(s, a) \in \mathcal{S} \times \mathcal{A}.$$

Time Homogeneous MDP
A time-homogeneous MDP is specified by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{R}, P)$.